Customer Shopping Behavior Analysis

Understanding purchasing patterns to drive sales performance and customer retention

Project Overview

Objective

Analyze how demographics, product categories, discounts, and seasonal trends influence customer decisions

Dataset

3,900 transactions with 18 features including demographics, purchase details, and behavior patterns

Tools

Python (pandas)

PostgreSQL (analysis)

Power BI (visualization)

Key Data Insights

Revenue by Gender

Male customers generate **$157,890** vs Female **$75,191**

Top Products

Blouse, Pants, Jewelry, Sunglasses lead sales across categories

Age Impact

Young adults contribute highest revenue at **$62,143**

Discount Strategy Analysis

50%

Hat Discount Rate

Most heavily discounted product

49%

Sneakers & Coat

High discount products

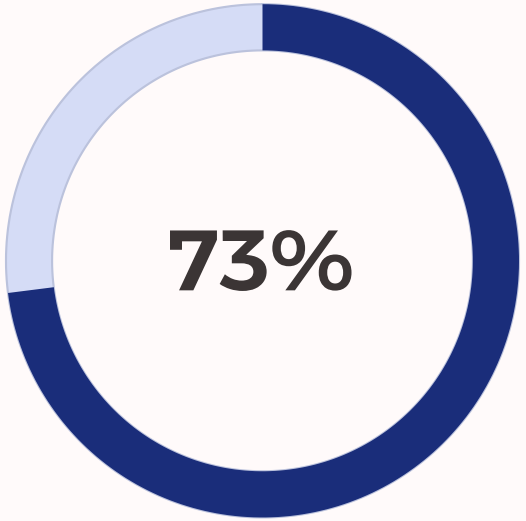
839

High-Spend Customers

Discounts effective even for premium shoppers

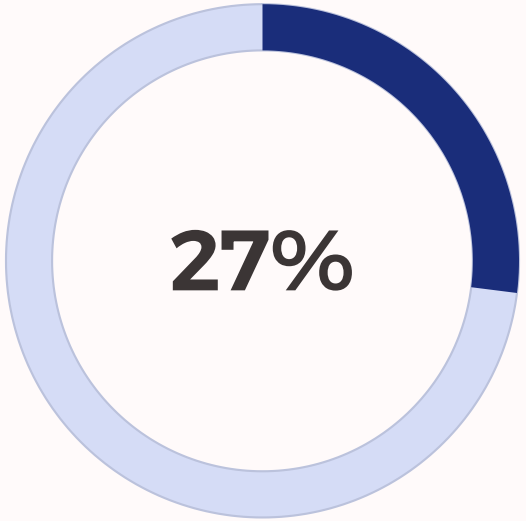
Discounts drive purchases across all customer segments

Customer Segmentation



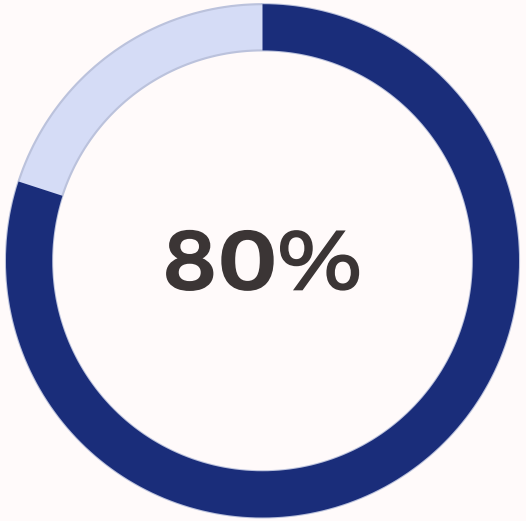
Non-Subscribers

2,518 repeat buyers without subscription



Subscribers

958 customers with active subscriptions



Loyal Customers

3,116 customers with strong retention

Payment & Shipping Analysis

Payment Methods

- Credit Card: highest revenue ($40,310)
- Debit Card: highest average spend ($60.92)

Shipping Preferences

- Standard shipping most popular
- Express shipping: higher spending ($60.48 avg)

Seasonal Performance

Fall

Best performer: $60,018 revenue

Spring

$58,679 revenue

Winter

$58,607 revenue

Summer

Lowest performance: $55,777

Product Category Performance

Clothing

$100K revenue, 1,700 sales

Accessories

$75K revenue, 1,200 sales

Footwear

$30K revenue, 600 sales

Outerwear

$15K revenue, 300 sales

Business Recommendations

01

Target Young Adults

Primary revenue drivers with highest spending patterns

02

Optimize Discounts

Focus on high-impact products like hats, sneakers, and coats

03

Enhance Subscription Value

Improve model to increase conversion from 27% to higher

04

Leverage Seasonal Trends

Launch campaigns around fall peak performance

05

Strengthen Retention

Build on strong loyal customer base of 3,116 customers

Key Takeaways

Male Majority

Generate 2x revenue vs female customers

Discount Effectiveness

Drives purchases across all customer segments

Strong Retention

80% loyal customers ensure stable revenue

Insights enable data-driven decisions to improve revenue and customer engagement